

Università
degli Studi
di Padova

Messa a punto di un potenziale interatomico per la simulazione della tantala amorfa tramite machine-learning

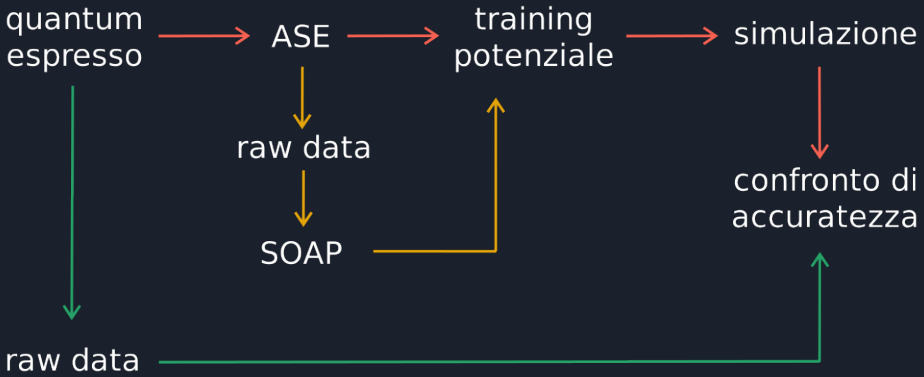
Relatore: Prof. Paolo Umari Laureando: Alberto Stocco

Anno Accademico 2025/2026

Indice

1. Struttura di lavoro
   - SOAP
   - Addestramento GAP
   - Simulazione
2. Python
3. Test sul dataset di QE
   - Energie e forze
4. Simulazione
   - NVE
   - NVE: Analisi
   - ZBL
   - Algoritmo Metropolis

Struttura di lavoro

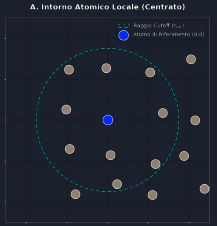


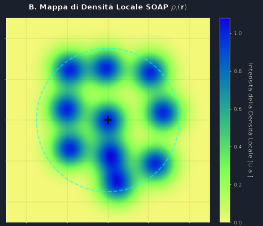
SOAP

SOAP: Smooth Overlap of Atomic Positions

Generazione della mappa di densità

$$\rho_Z(\vec{r}) = \sum_{\alpha}^{Z} e^{-\frac{|\vec{r}-\vec{r}_\alpha|^2}{2\sigma^2}} \qquad (1)$$





SOAP

Espansione in funzioni di base

$$\rho_Z(\vec{r}) = \sum_{n=1}^{n_{max}} \sum_{l=0}^{l_{max}} \sum_{m=-l}^{l} c_Z(n,l,m) g(r\,;n) Y(\theta,\phi\,;l,m) \qquad (2)$$

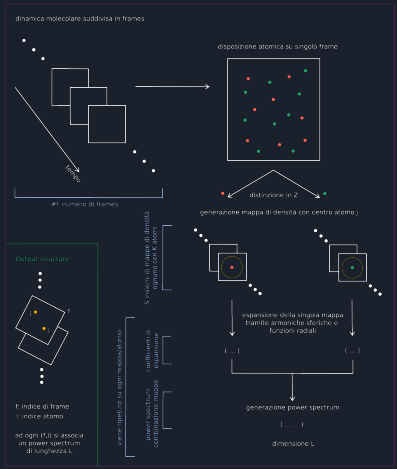
$$c_Z(n,l,m) = \int_{\mathbb{R}^3} \mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\phi \;\; g(r\,;n) Y(\theta,\phi\,;l,m) \rho_Z(\vec{r}) \qquad (3)$$

Calcolo del power spectrum

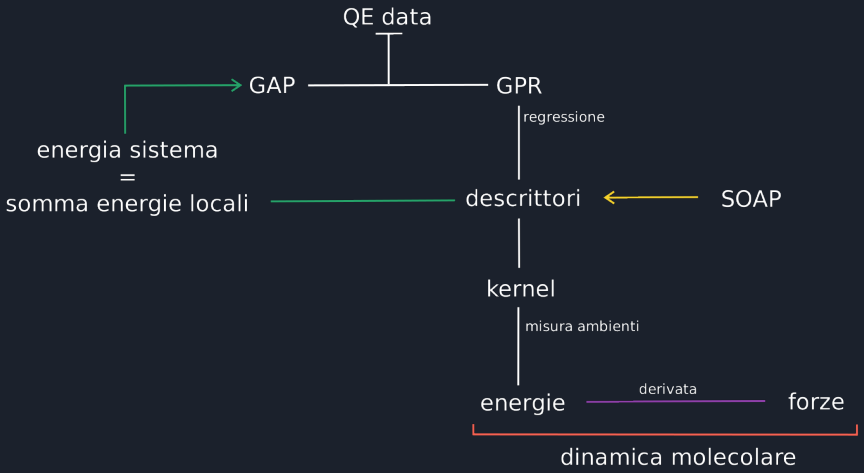
$$p_{Z_1 Z_2}(n,n',l) = \pi \sqrt{\frac{8}{2l+1}} \sum_{m=-l}^{l} \overline{c_{Z_1}}(n,l,m) c_{Z_2}(n',l,m) \qquad (4)$$

Risultato finale

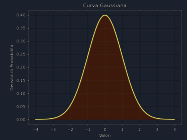
$$\dim = (l_{max}+1)\,\frac{1}{2}\,(S \cdot n_{max})(S \cdot n_{max}+1) \qquad (5)$$



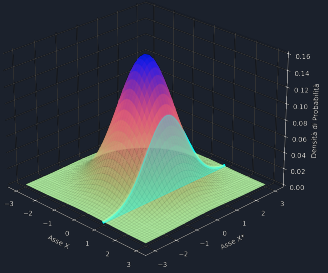
GAP


QE data
GAP
GPR
regressione
energia sistema
=
somma energie locali
descrittori
SOAP
kernel
misura ambienti
energie
derivata
forze
dinamica molecolare

GPR



Gaussiana bivariata



Simulazione NVE

Simulazione dove si impone la conservazione di:

- numero di particelle
- volume
- energia

ZBL

ZBL: Potenziale Ziegler-Biersack-Littmark

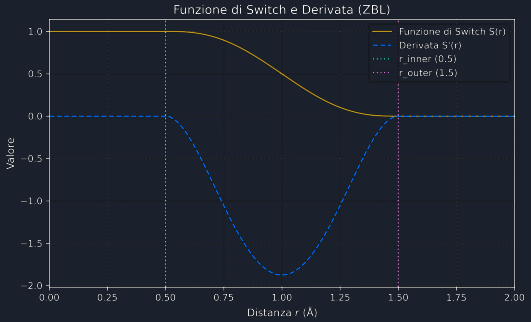
$$V(r_{ij}) = \frac{1}{4\pi\epsilon_0}\frac{Z_i Z_j}{r_{ij}}\phi\left(\frac{r_{ij}}{a_{ij}}\right) \tag{6}$$

$$\phi\left(\frac{r_{ij}}{a_{ij}}\right) = \phi(x) = 0.1818e^{-3.2x} + 0.5099e^{-0.9423x} + \\ + 0.2802e^{-0.4029x} + 0.02817e^{-0.2016x} \tag{7}$$

$$a_{ij} = \frac{0.8854\, a_0}{Z_i^{0.23} + Z_j^{0.23}} \tag{8}$$

$$S(u) = -6u^5 + 15u^4 - 10u^3 + 1 \tag{9}$$

$$u = \frac{r - r_{inner}}{r_{outer} - r_{inner}} \tag{10}$$

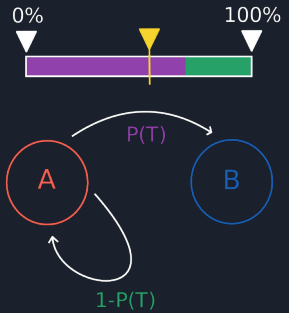


Simulazione con algoritmo di Metropolis

$$P(A) \propto \exp\left(-\frac{E_{pot}(A)}{k_b\,T}\right) \tag{11}$$

$$\Delta E = E_{new} - E_{old} \tag{12}$$

$$P_{acc} = \exp\left(-\frac{\Delta E}{k_b\,T}\right) \tag{13}$$



Titolo della Slide

L'analisi è stata svolta in python usando alcune librerie

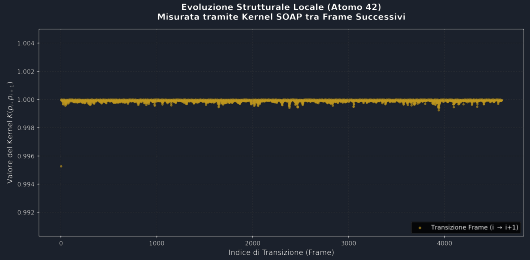
- ASE
- DScribe
- Quippy

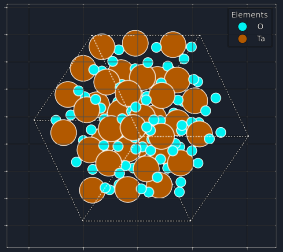
Dataset A

Università degli Studi di Padova

Divisione dei dati: Training set Test set

Verifica dei risultati soap



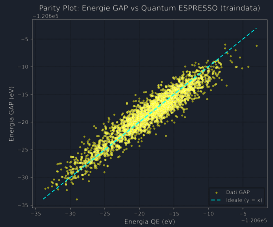


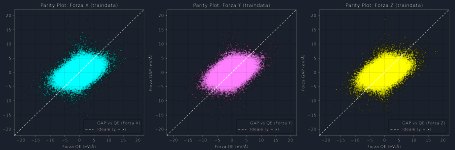
Frame del dataset A generato con matplotlib

Verifica delle energie e delle forze

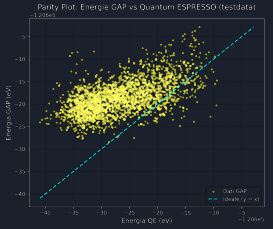
Università degli Studi di Padova

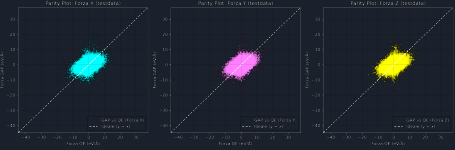
Training set



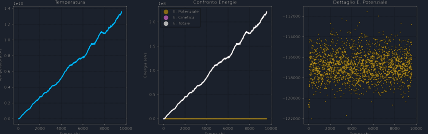


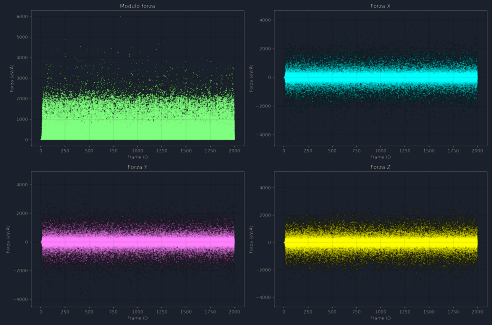
Test set



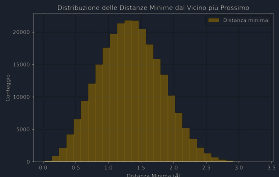


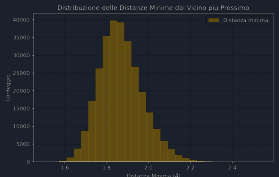
Simulazione NVE

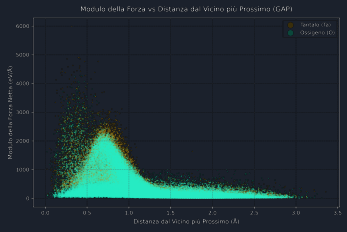


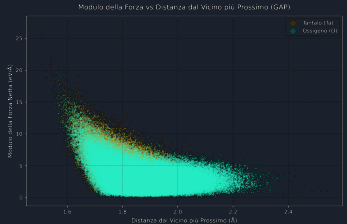


Simulazione NVE: analisi



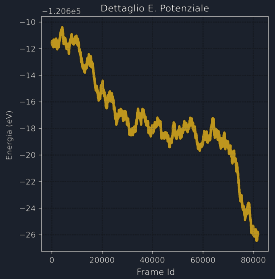


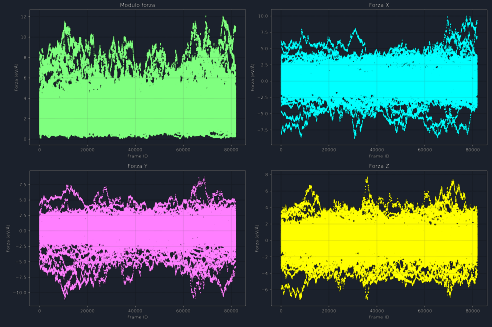




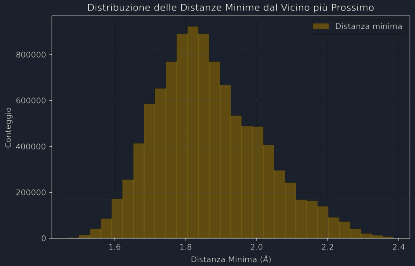
ZBL: risultati

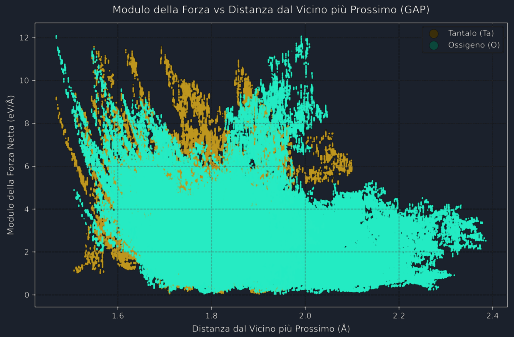
Algoritmo di Metropolis: risultati





Algoritmo di Metropolis: risultati





Domande??